



Enterprise Asset Management (EAMS) v15

Mobile Work Order

End User Documentation

Update 3/1/2024

MaintStar Mobile

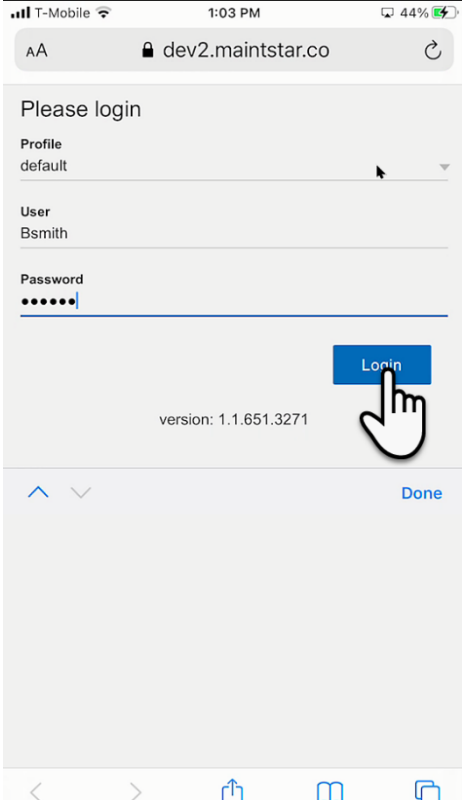
MaintStar Mobile is a web-based application designed for you to manage your work orders on a phone or tablet when you cannot be at your desktop. It can be used on an Apple or Android device. A tablet gives you more screen space making data and maps easier to see and use. A tablet will not fit in your pocket but can be carried like a book. A tablet will also have a larger battery so the device can go longer between charges. A phone is of course smaller. The smaller screen makes text smaller on the screen to read and buttons more difficult to click precisely. You trade these shortcomings in a phone for the convenience of a device you can put in your pocket. If you are not tech savvy and have trouble clicking on a screen then the tablet will probably be easier for you. Whichever device you choose, the MaintStar mobile application will work with you throughout the day. We will use an Apple iPhone 8+ with the default Safari browser in this example. We will also show what the app looks like on a tablet for comparison.

The MaintStar Mobile Work Order will allow remote workers to access their assigned work order in the field. Key features of the Mobile work order are:

- The Mobile Work Order is easy to use.
- Users can view their work orders on Android and iOS tablets and phones through a cellular connection.
- The system will leverage your GIS map services
 - You can view work orders on the G.I.S. map.
 - You can add assets to the work order from your GIS map layers or from searchable drop downs.
 - You can view asset history from the map.
 - You can markup the map to show work areas.
- We seamlessly integrate your camera to take pictures and add them to the Work Order.
- Users will be able to add work detail, hours, parts and materials, contractor information and other closing information.
- They will be able to close the work orders in the field.
- If they will be traveling to areas where cell reception is spotty, the system will allow them to download their work orders to the device while they have cell service. They will also be able to download selected assets, and work in 'Disconnected' mode when they travel to no reception areas. When they later reenter cell service area, their device will sync back to the server.

Enter Your User Id and Password and Select your Work Group
Enter your Id and password

Select the 'Profile' you want to log into from the drop down. You may have only one profile, or you will have more than one if you have a production and a test environment for example. When you enter your ID and password on a phone or tablet, be aware that most devices will try to capitalize the first letter. If your password is all lower case, you will need to shift to lower case before you type the first letter. Press 'Login". You will be taken to the Home Screen.



Allow your device to use your location

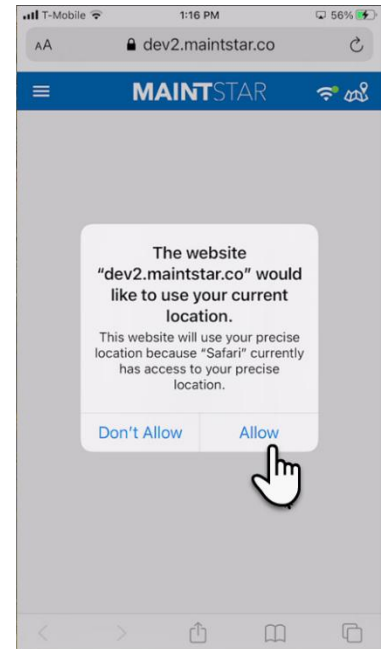
Your device will ask you if you want to use your current location. **You should always allow this.** Using the location services of the device will allow the map to know where you are geographically. If you accidentally click don't allow, you should close the app and start it again as this is the only time the location service can be activated.

App and data will update and sync. This may take a minute or two depending on the size of the database.

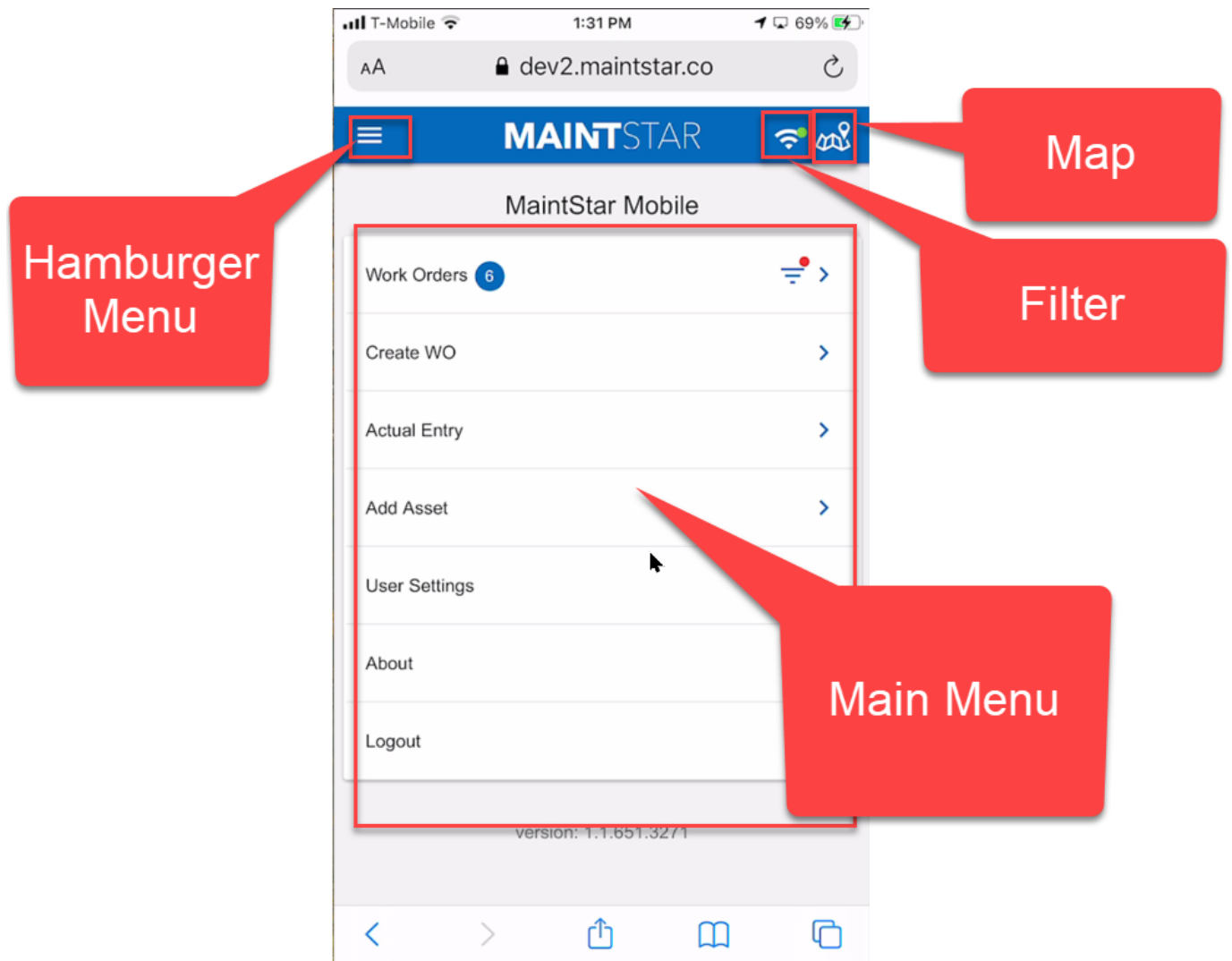
You can manually update the data manually if needed.

The app will take you to the last screen you were on when you closed out last time. It will download all the open work orders assigned to you.

In this example we are starting with an overview of the screens.



Main Menu Screen



Across the Top Blue bar, the three icons are

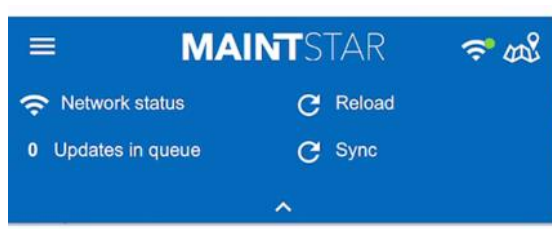
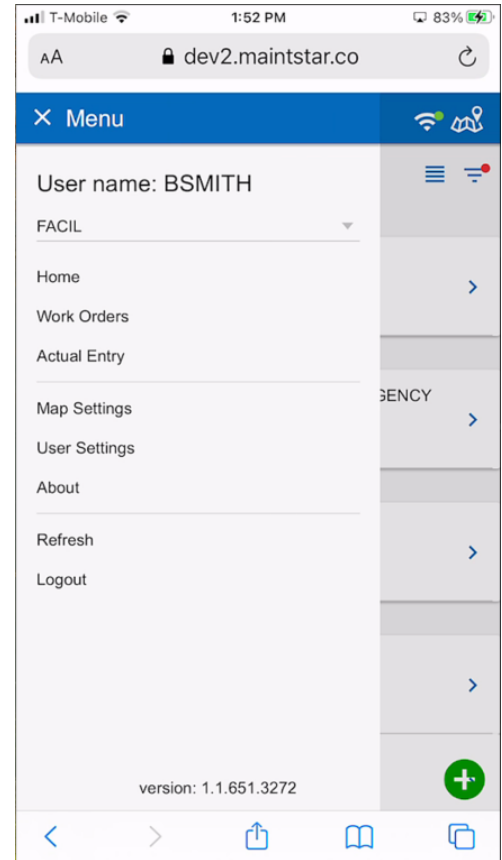
- "Hamburger menu"
- Filter
- Map

The Large Main Menu section

- **Work Orders** takes you to the work orders section. There you can see work orders in a table view and on the map. You can edit them and create new work orders. This will be the main section that users will work in. The Work Orders menu shows a number in a circle. That is the number of work orders downloaded for you when you logged on.
- Work orders assigned to you will be loaded when you log on. The system looks for your ID in the
 - 'Assigned to field',
 - The Crew Leader Field,'
 - The supervisor field and as
 - 'Labor' on the LEM section.
- **Create Work Order** is a direct link to a new work order entry. This is available from the Work order section with a green circle with a plus sign in it also.
- **Work Planning Entry** allows you to create work plan entries that are either not connected to a work order, and ones connected to a work order. If the entries for time and materials etc. **are** going to a work order, you should add then to the work order directly as there will likely be other detail for you to fill in on the work order beside LEM.
- User settings shows information about your ID. It gives you a selection of a light or dark theme and the selection to pull data on-line or Pre-load them.
- About the app shows version number and contact info.
- Log out is how you log off the application.

The Hamburger Menu in the upper left corner, opens the following slide out screen:

- It shows your log on ID.
- You will see the Work group you are in. If you have access to different workgroups you can change to a different workgroup here, in the drop down.
- Other menu items are the same as on the home screen except for the refresh.
- This icon in the blue header bar  is Refresh. Click it.
- The menu item 'Refresh' will open the refresh panel.



The green dot shows that your data is synced, meaning what you have entered on your device has been uploaded back to the server.

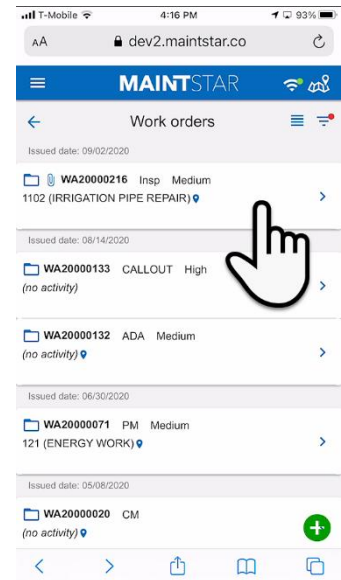
You can reload your work orders if necessary, with the reload button. Sync will sync your entries to the server manually. Normally the system will sync date automatically.

Work Orders

Work orders assigned to you will be loaded when you log on. The system looks for your ID in the

- 'Assigned to' field,
- The 'Crew Leader' Field,'
- The 'supervisor' field or as
- 'Labor' on the LEM section.

When you click on this menu item the list of work orders will open.

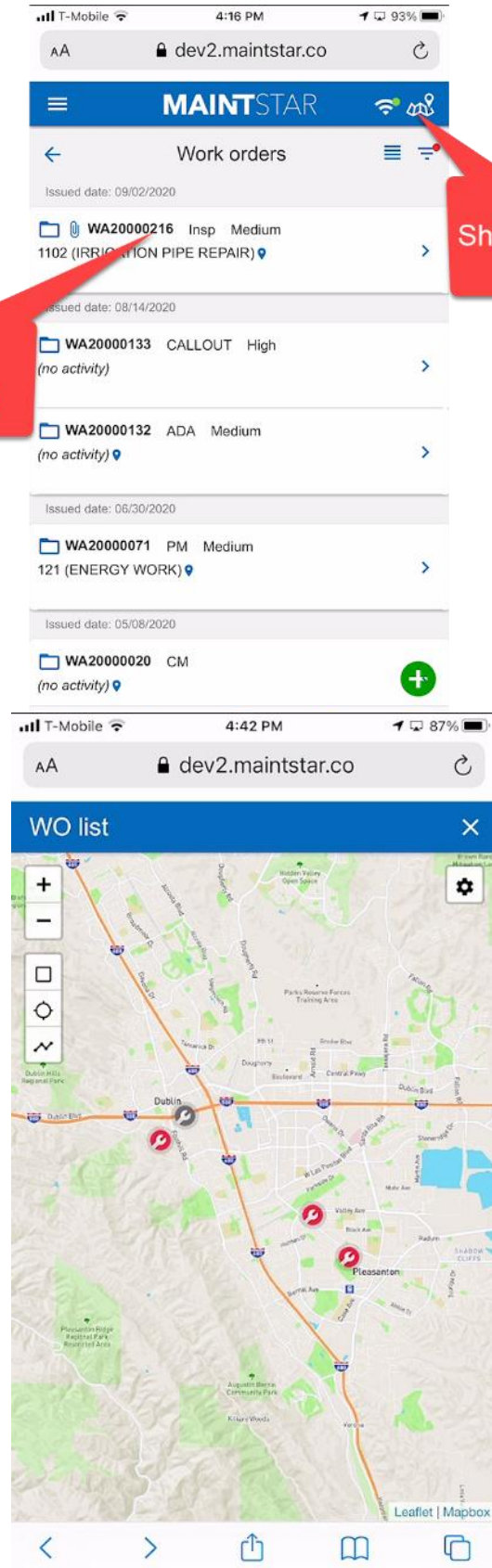


The Map Icon will open the Map screen.

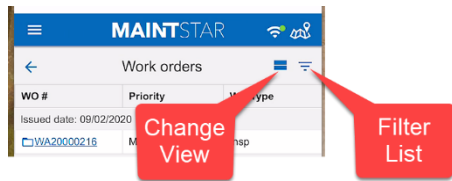
If the work orders have GIS Assets in them, then the first asset in each one will be shown on the map with a wrench icon. If you tap on one of the icons the id of the asset will be shown. Open Work Orders will be red Closed ones will be Gray. You can pinch to zoom in and out. Close the Map with the left x in the corner to go back to the work order list.

Work Orders
(Click to open)

Show on Map



The Work Order List Views



The work order list has two icons in the upper right corner. The one on the left changes the list view and the one on the right is used to filter the list.

There are two ways to see the list of work orders. They can be shown in a condensed list view or in a more expanded list view. The default is the expanded view. Choose which ever display works best for you.

The condensed list view shows the work orders in a scrollable table, 10 items to a page. Each work order takes up just one line. The condensed list view can be scrolled left to right to see more information. Clicking on the work order number opens the specific work order.

Expanded list view

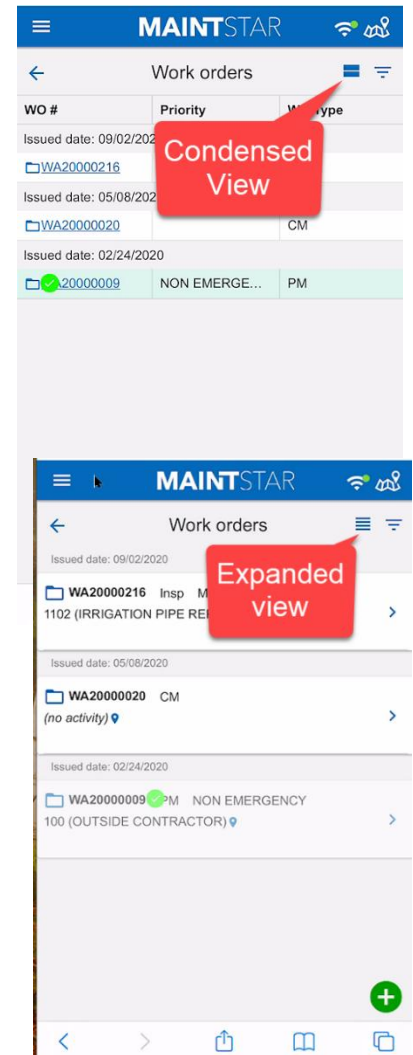
The expanded list view displays the work orders according to the settings in the filter and grouping. The system will remember the way you had them set last and opens the way you left them. Closed work orders have a solid folder icon, open ones, white.

A paper clip indicates an attachment.

The pinpoint teardrop indicates one or more mapped asset.

The expanded view shows you a few essential things about the work order in a small card without scrolling left or right.

Click the right pointing arrow on the right side of the card for the work order you want to open in detail.



The Filter

The filter has two functions, to filter the work orders that are displayed and a control the grouping. The tabs at the top are used to switch between these two functions.

Wo #, status, priority and assigned to, are choices. If you click 'Show More' you will see many more fields to select from.

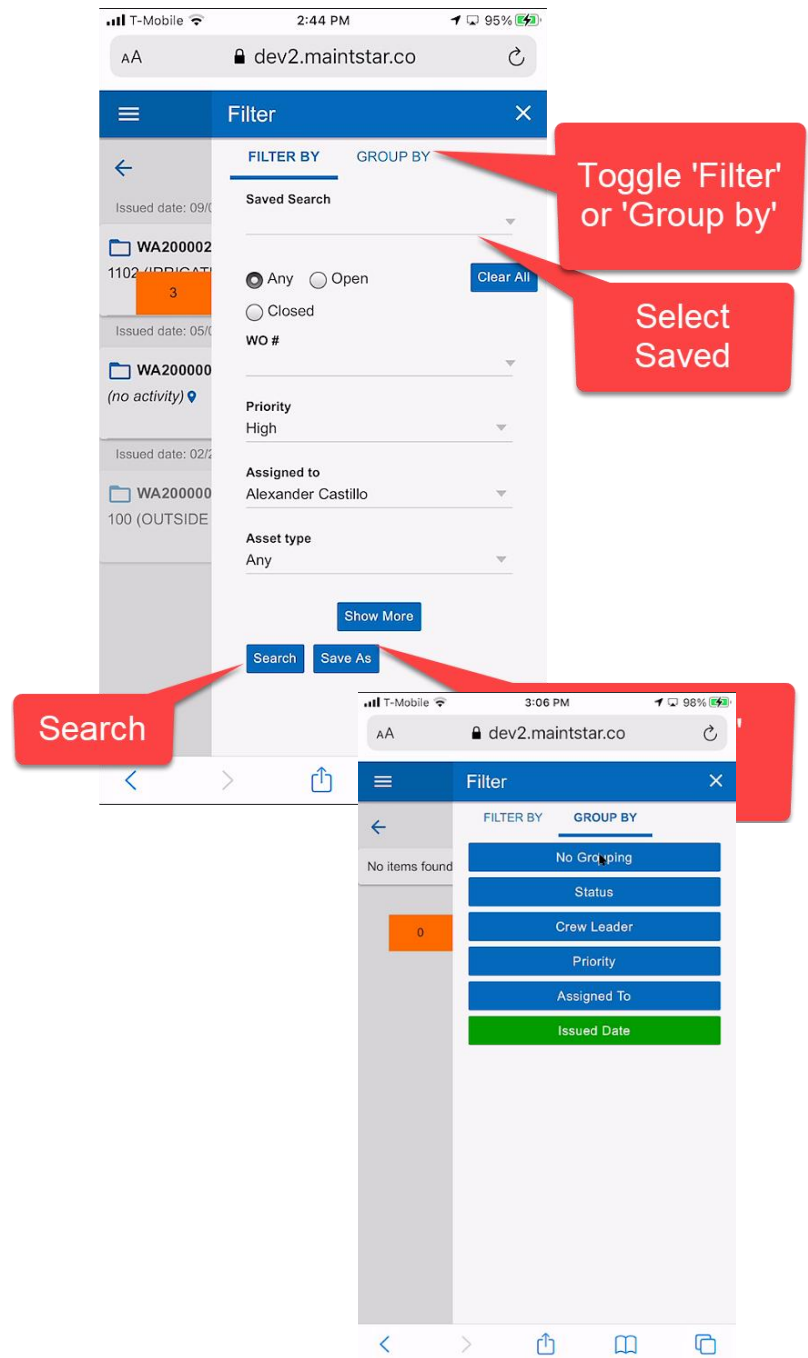
Click 'Search' to change the displayed Work orders to ones that meet this selection.

'Save as' is useful if you have a filter selection you want to use many times. You can give it a name and find it in the 'Saved search' drop down.

'Clear all' resets the filter.

Group By has several choices to group your results. You can select

- No Grouping,
- Grouping By:
 - Status,
 - Crew Leader ID,
 - Priority,
 - Assigned to ID, or by
 - Issue date.



Work Order Screen

Click on a work order number to open the work order in detail.

Across the top of the Work Order screen are four icons.

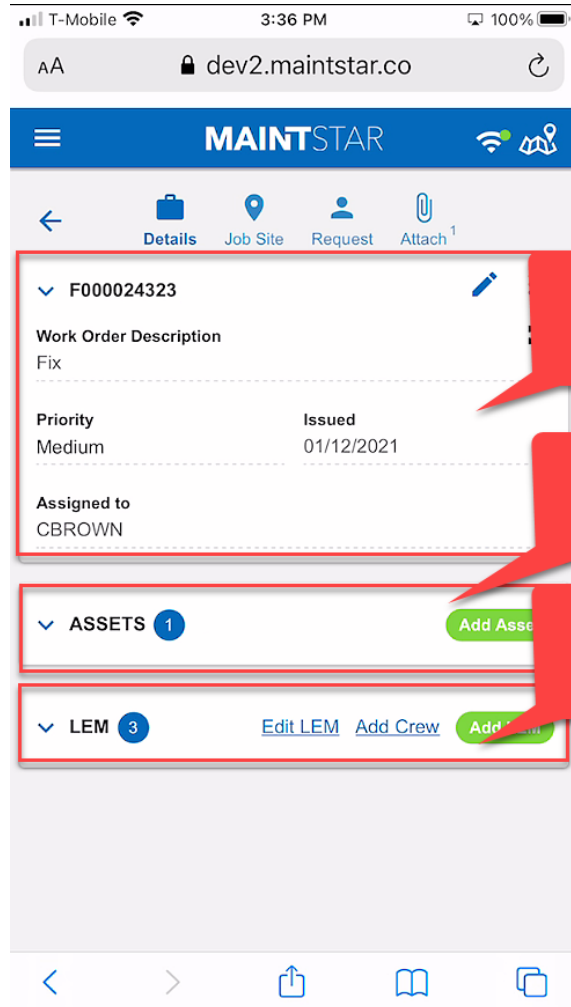
- **Details** - is the main Work Order Information view
- **Job Site** - is a place you can record an address of where the work is done.
- **Request info** – If the work order came from a request, the requestors contact information will be here.
- **Attach** -This is where photos and other attachments are stored. Yes, you can take pictures with your phone or tablet and attach them to the Work Order! You will see a number indicating how many attachments there are.

The **Work Order Detail** screen is divided into sections. The example below shows these panels in their condensed state. Each one has a small arrow in the upper left. These arrows allow you to open a section up or condense it.

The arrow head points down when the section is condensed, and it points up when it is expanded. The top section is the header. It holds general information about the work order.

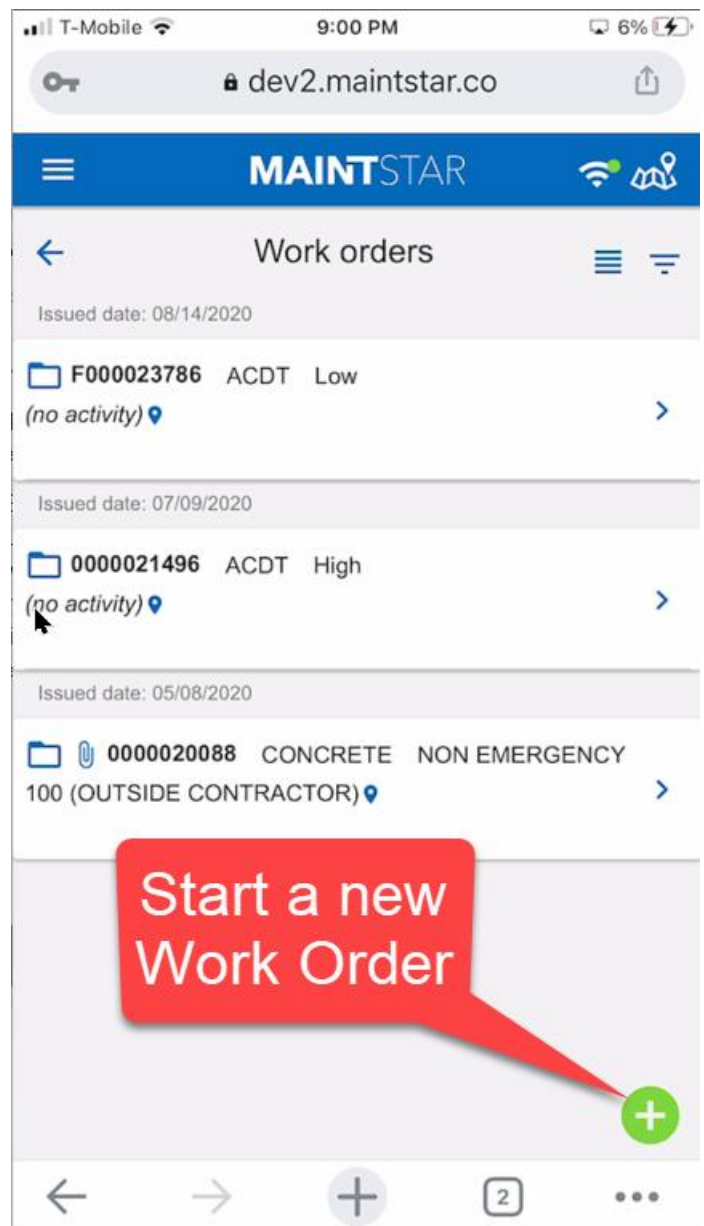
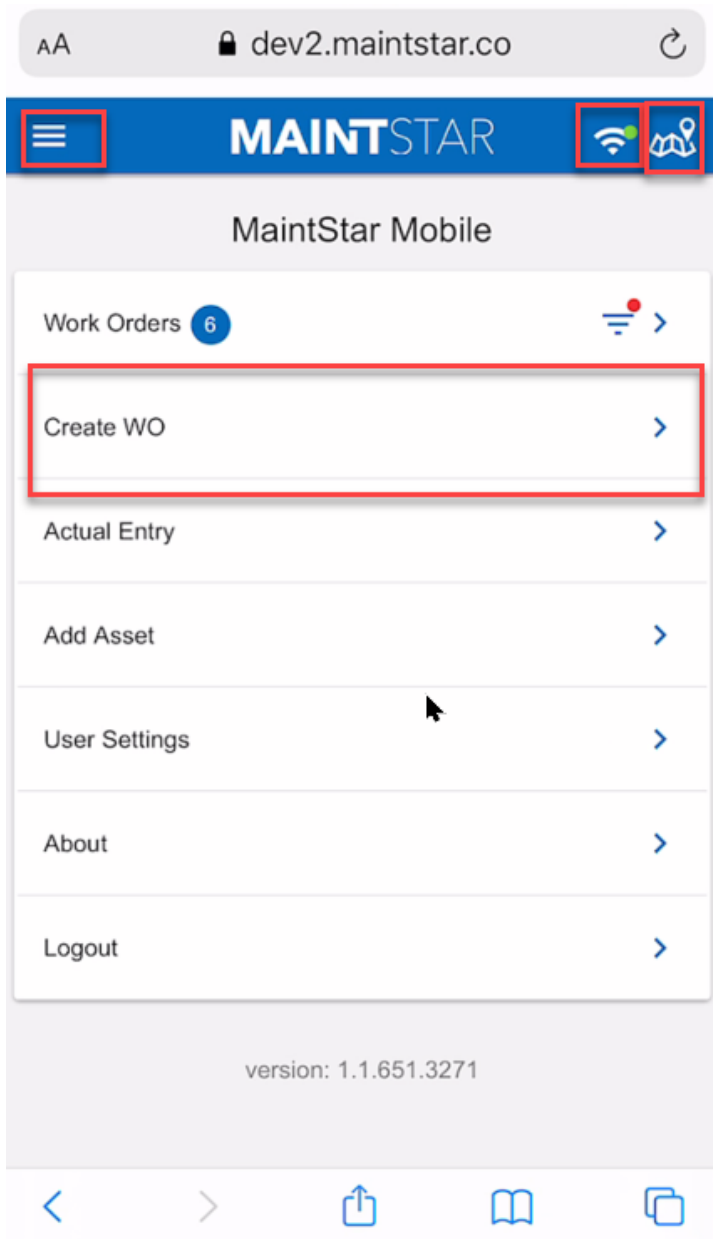
Below that is the section for assets.

Next is the LEM section where Labor, Equipment Materials and Work Quantity are entered.



Starting a new Work Order

You can start a new Work order from the main menu “Create Work order or with the green plus sign on any existing work order.



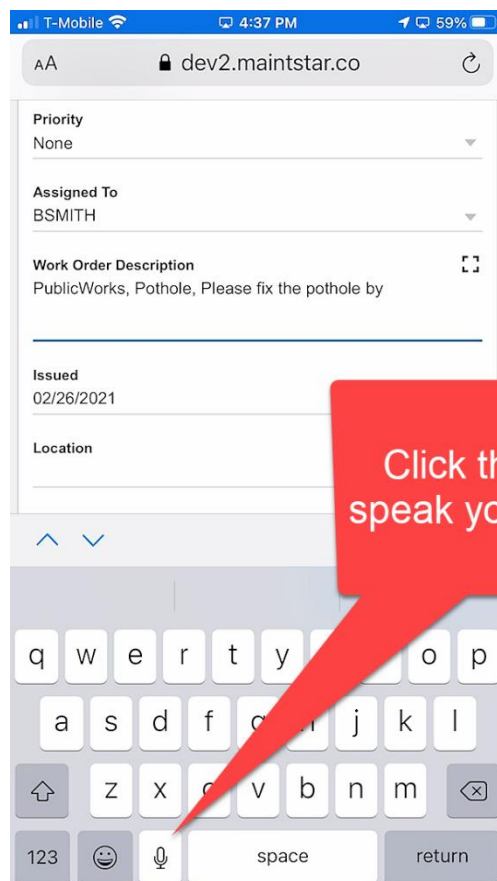
Filling in the New Work Order – Header section

The new work order will be started. You will see the header section. Complete the fields such as work order type, add priority and assignments for assignee, crew leader and supervisor if you use these designations. Enter a short description of the work to be performed in the wo description field.

Click Save at the bottom of the screen. You can now condense the header section if you wish with the up arrow at the top of the header section. This is up to you. There is nothing wrong with leaving it expanded, you will just have to scroll to get to the asset section below it. We would however suggest that if you expand only one section at a time, you might find it easier to navigate and less confusing.

You can talk your entries into text fields

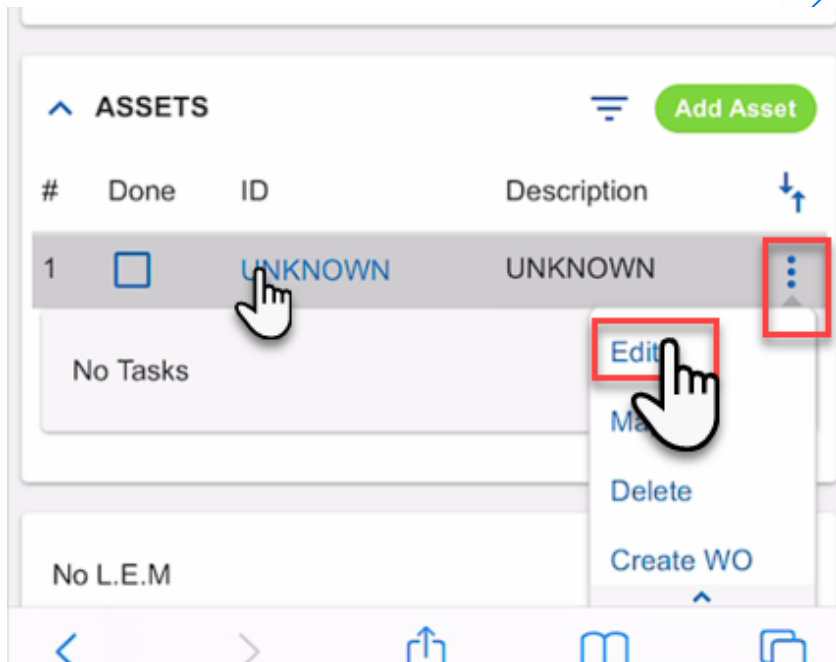
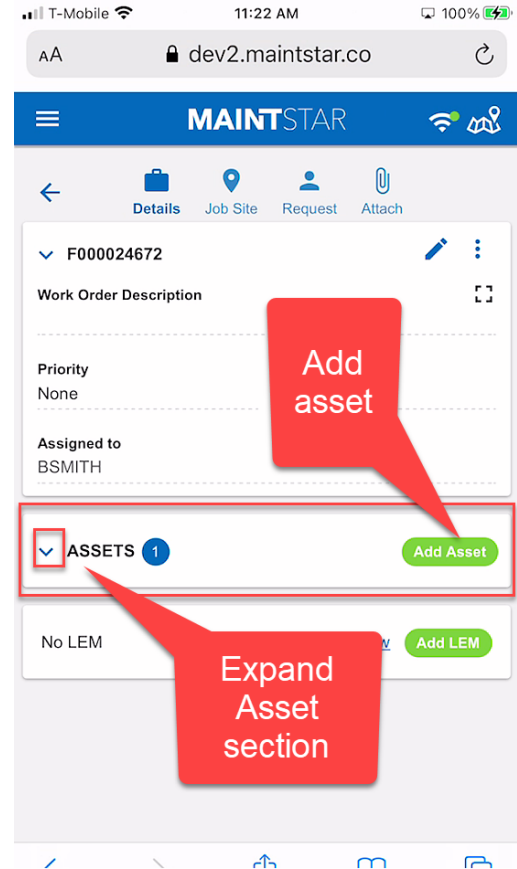
Phones and tablets have great speech to text capabilities. When you click into a text field the keyboard will pop up. If you click the microphone, you can talk your entry in and have almost no typing necessary!



The Asset Section – Standard Layout

The asset section can be opened with the up/down arrow on the left. If your work group is set up for the **Standard Work Order Form**, there will be one asset added by default.

It has the ID of “Unknown” In the Standard work order format each asset has its own Labor, Equipment and Material. You can add labor to the Unknown asset. It is however intended that once you begin the service you will at some point determine the asset involved and change the Id from Unknown to the correct Asset ID. To change the ID from ‘Unknown’ to an actual asset Click the 3 dots on the right of the asset ID. In the drop down choose ‘edit.’



You will go to a search screen.

Here you can select the asset you need from the asset drop down or search for it with specific fields. To search click the magnifying glass. In the search screen put in one or more criteria then click search.

Select from the Search Results

You will see a list of matches for your criteria. Tap on the right one to select. It will be added to your work order.

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Edit Asset UNKNOWN

SEARCH CRITERIA RESULT 0

Saved Search

On-line Pre-loaded Clear All

Search

Asset ID Asset Description

Asset Type Building

Location

Sub Location

Street Name Block Num

Quick loc ref Neighborhood

Put in search criteria, then click search.

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Edit Asset UNKNOWN

SEARCH CRITERIA RESULT 6

Saved Search

Search

Asset ID	Asset Type	As De
BUILDING - TEST	Building	BL
Covid19 Hall of F...	Building	Cc
Floor-01	Building	Fk
Main Library	Building	Mi
Room-01	Building	Rc
Village Hall	Building	Vil

Asset is added

MAINTSTAR

VICTOR

ASSETS

#	Done	ID	Description
1	☐	Village Hall	Village Hall

No Tasks Add Task

No L.E.M Add LEM

No Attachments Add Attach

Asset Section – Infrastructure Layout

If your workgroup has been configured for the Infrastructure layout. Your asset section will be a little different than the Standard layout design. The Assets will be in a table where multiple assets can be listed. The assets will have ONE section for LEM, Labor, Equipment and Materials. This means that the work order is capturing the cost for “the entire” work order as a whole job. The Standard Work Order Layout has a separate LEM section for each asset.

In the following example we will switch to a water work order and add some hydrants from the GIS map instead of the asset search.

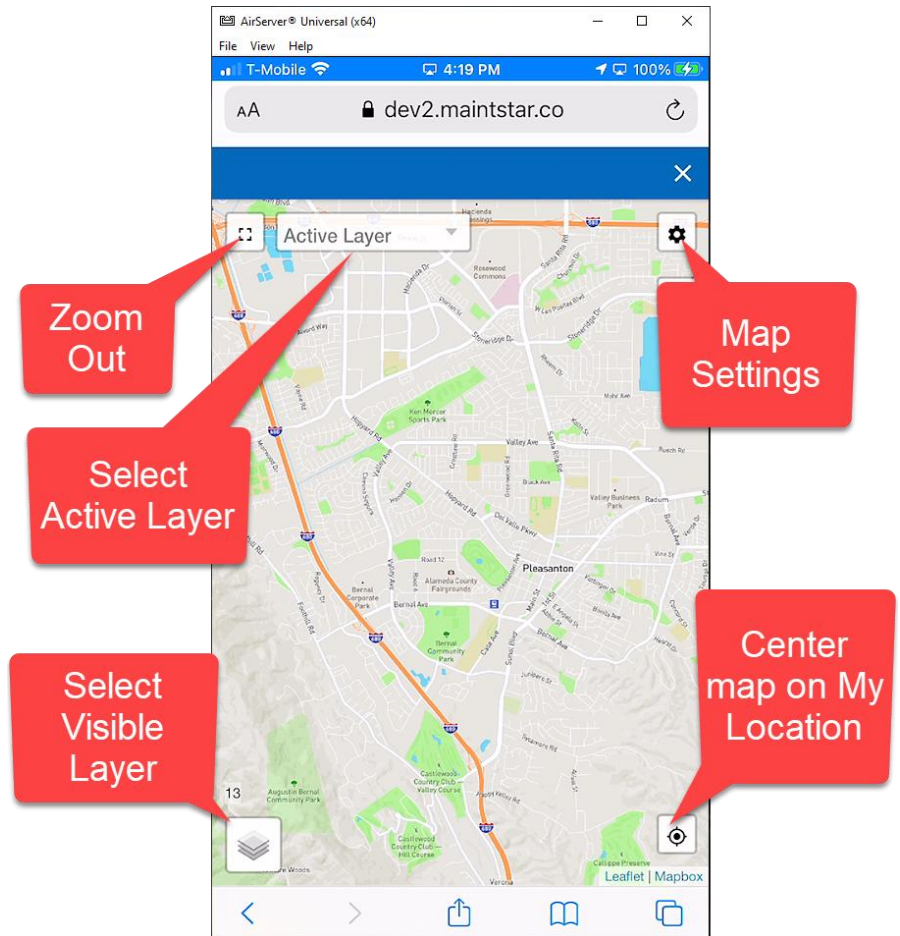
Adding Assets from the Map

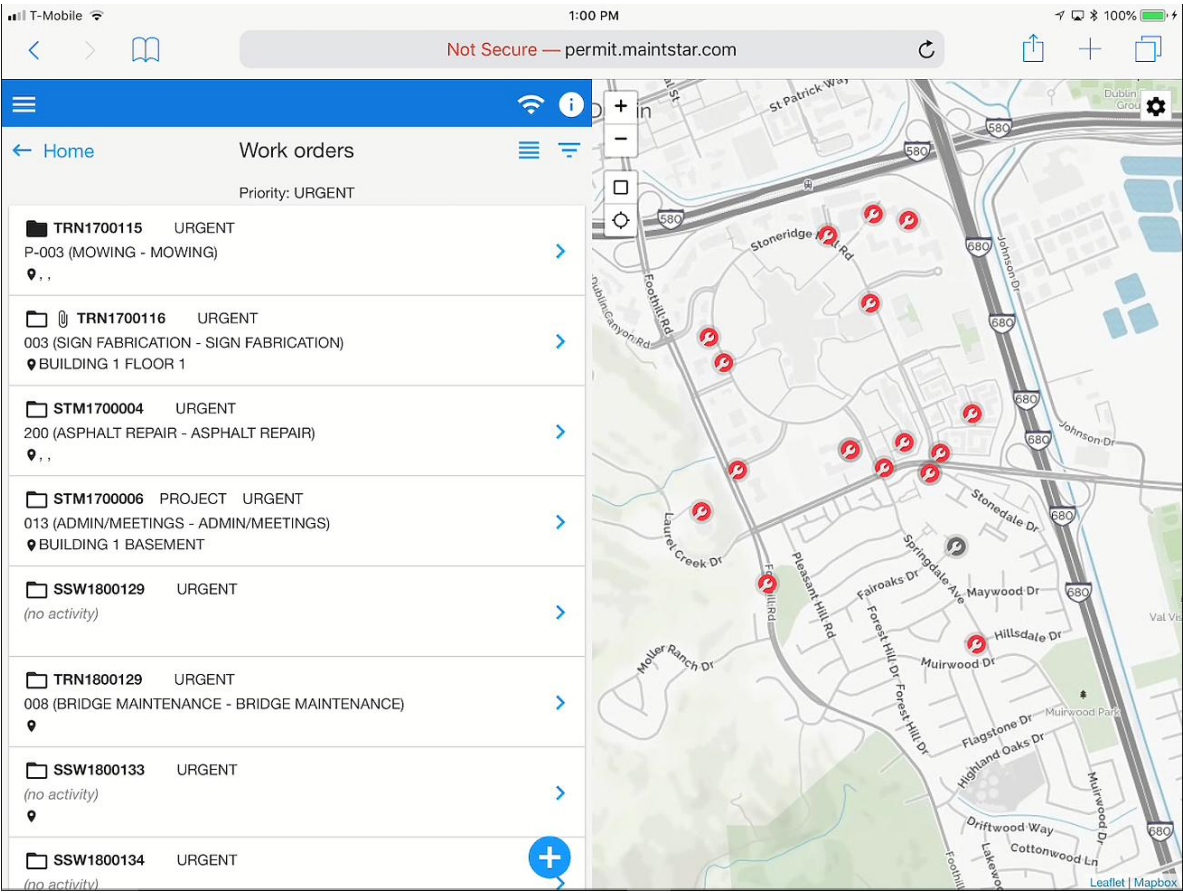
Your GIS assets can be selected from the map and added to your work order. To do this, start a new work order with the green plus button at the bottom of the work order screen. Then open the Map with the map icon in the upper right corner of the screen.

The Map will appear full screen. Note when you want to go back to the work order, use the blue back arrow at the bottom of the screen or the white X in the upper right corner.

Note: If you are using a tablet you have the advantage of a screen that is big enough to be able to show the work order and the map at the same time. On a phone, you must open the map to see the map then close it to go back to the work order screen.

TABLET VIEW – See the map and Work Order side by side.





Click the configuration Icon which looks like a gear.

The configuration has two radio buttons labeled On-Line and Pre-Loaded.

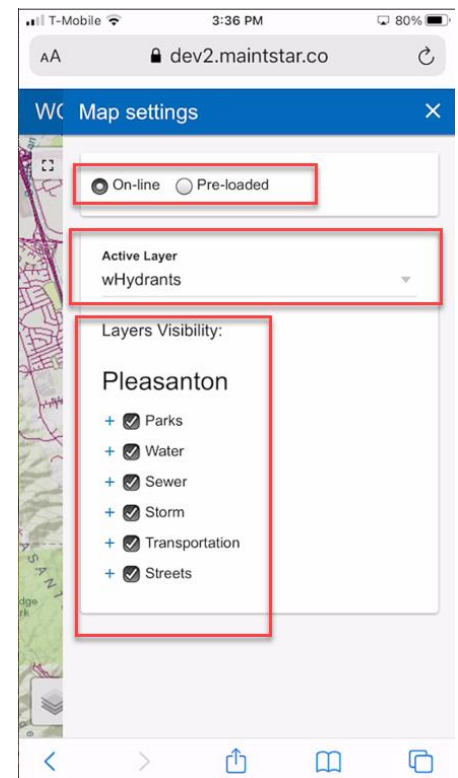
The On-Line selection will pull assets to the map over your internet connection. This is the normal way to work. If you anticipate working in an area where there is little or no internet connection you can use the pre-loaded function to download asset types you will be working with, while you have internet connection. They will be stored on the device memory. Because of this, you will be able to work with these assets and add them to work orders where you do not have good internet connection.

In this example we will use the pre-loaded connection method.

The Active Layer dropdown needs to be set for the asset type that you want to select on the map. Here we have it set for 'wHydrants', meaning water hydrants.

You can have many layers visible in the Layers Visibility section, but you must choose which layer to select from in the Active layer.

When the active layer is set, close the panel by swiping it right



You will be taken back to the map screen. You will see the work Order that you are on at the top. Below that is the active layer listed at the top of the screen. If you are at the location, you are going to be working at you can tap the Center Icon in the lower right. The screen will zoom to your location. You must have location services turned on and active in the device for this to work. If you are not there you can pinch and zoom the map to move to the location, you need. You will need to zoom in close to see the icons for the hydrants. In a GIS map, fine detail such as asset icons do not appear when you are zoomed at a high elevation. Select the asset by clicking on it. It will then be in the box at the bottom of the screen.

If you click it there you will see it highlighted in green on the map to make sure you have the right one.

The x will remove the asset from the selection box. The plus sign will add it to the work order.

If you added it by mistake, click the plus sign again. You will see a green check indicating that it has been added. Click the plus sign again at it will show a garbage can. Click the garbage can to delete the asset.

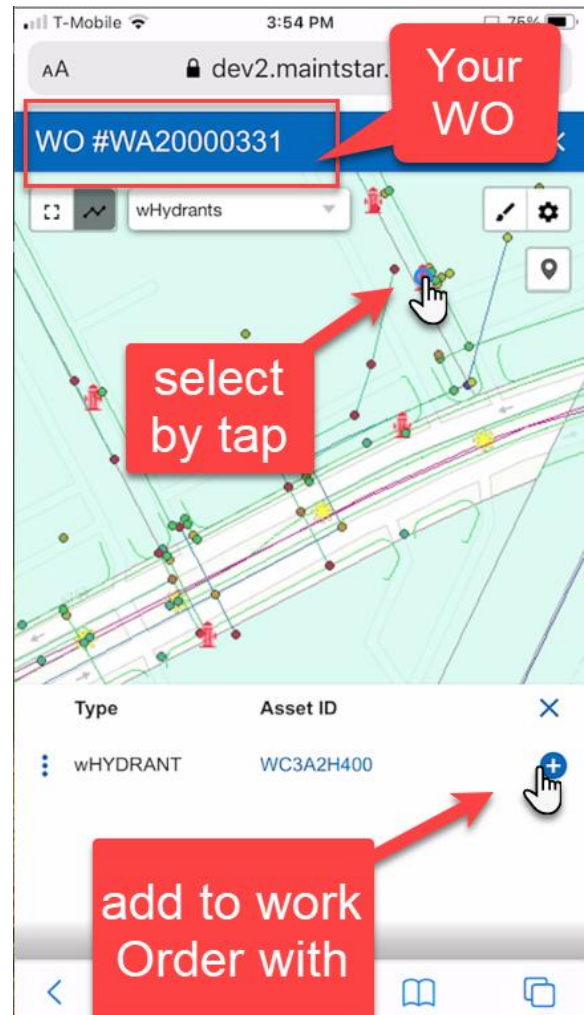
When it is added the Hydrant Icon will display a number 1, meaning the first asset added to the work order. If you need to add more than one hydrant, keep the map screen open and select other hydrants in the same manner. They will each be added to the work order with the plus sign. You can come back to the map from the work order to add assets again.

Working with Assets

All assets that are on the work order (which came from the map) are shown in the map view for the work order you have opened.

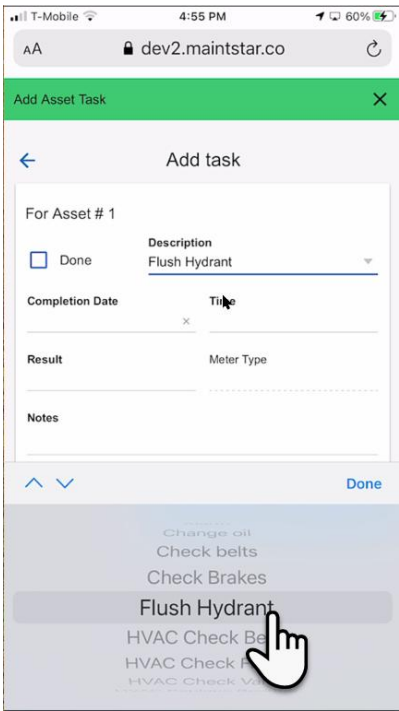
NEW CONCEPT: In version 15, each asset has a check box labeled 'Done'. This check box can be checked with a green check indicating that this asset is finished. The corresponding dot on the map will change from a blue dot to a green dot to show that it is done.

The +/- sign to the right of the check box expands and contracts the asset line itself for one asset.



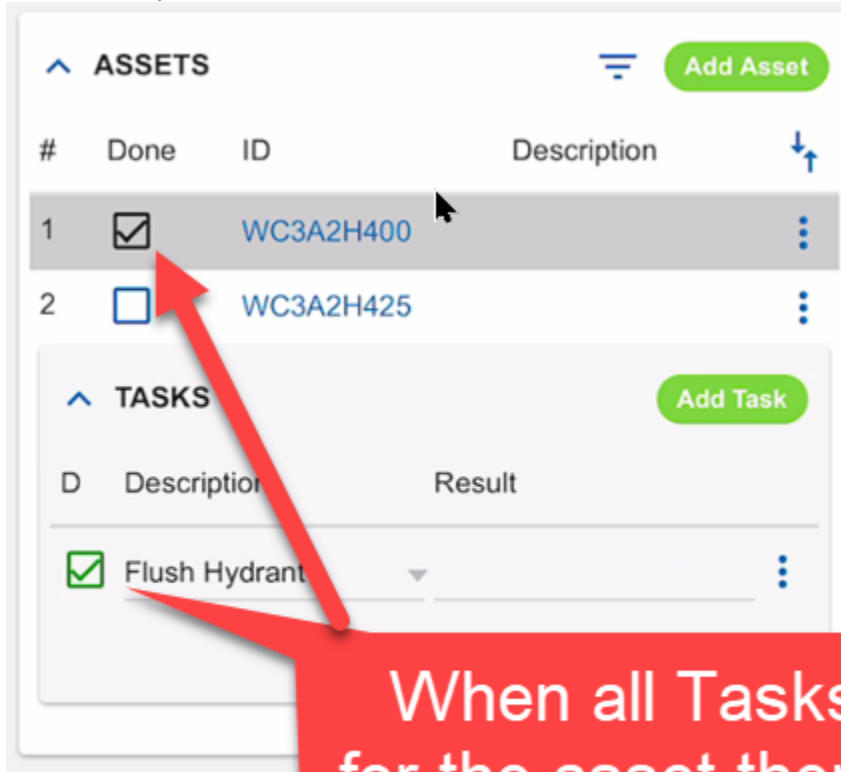
NEW CONCEPT: When the asset is expanded there is a new feature called 'Predefined Task'.

This feature will allow you to do many things:
Tasks will allow you to create a list of specific tasks, and the user can mark off which tasks are completed and when they were completed.
Tasks can also be set up for recording information, observations, conditions, and measurements.



Completing the Task

When a task is completed you should check the done checkbox on the task. You can add notes or other comments if needed. The system will automatically record the date and time the task was done. Once all tasks for that Asset are marked done, the system will check the asset Done box for you automatically.



When all Tasks are done
for the asset then the Asset
is marked done.

L.E.M.

LEM stands for Labor, Equipment and Materials. L.E.M. It is another section of the work order after assets. It can be expanded and condensed like the other sections.

To view the labor and equipment and materials expand the section.

You will see a list of all the Equipment Materials and labor currently on the work order.

- The category of each one is the first column.
- The second column is the identity of each entry. This column can be expanded by clicking on the ID.
- The third column is the quantity as a count or hours.
- The last column is the date

Click on a LEM entry to open it in full screen detail. There you can enter hours, pay type, Management Unit and Activity.

The screenshot displays the MAINTSTAR mobile application interface. At the top, the status bar shows 'T-Mobile', signal strength, time '5:49 PM', and battery level '84%'. The app header includes the MAINTSTAR logo and a user icon. Below the header, the work order ID 'WC3A2H425' is visible. The 'TASKS' section is expanded, showing a table with columns 'D', 'Description', and 'Result'. A task 'Flush Hydrant' is listed with a green checkmark. Below the tasks, the 'L.E.M.' section is expanded, showing a table with columns 'Type', 'Name', 'Qty/Time', and a date column. A hand icon points to the 'Add LEM' button in the top right corner of the L.E.M. section. The table contains four entries: LBR for Blaine Smith (5.5 hrs), LBR for Brett Kolve (4 hrs), EQP for 12457 (4 hrs), and MAT for 6" HYDRA... (5). At the bottom, the 'ATTACHMENTS' section is visible with a count of 1 and an 'Add Attach' button.

Type	Name	Qty/Time	
LBR	Blaine Smith	5.5 hrs	09/29/2020
LBR	Brett Kolve	4 hrs	09/29/2020
EQP	12457	4 hrs	09/29/2020
MAT	6" HYDRA...	5	09/29/2020

2 ways to display LEM Entries

In the setup of the system on the desktop there is a switch to control how LEM entries are entered on the Mobile.

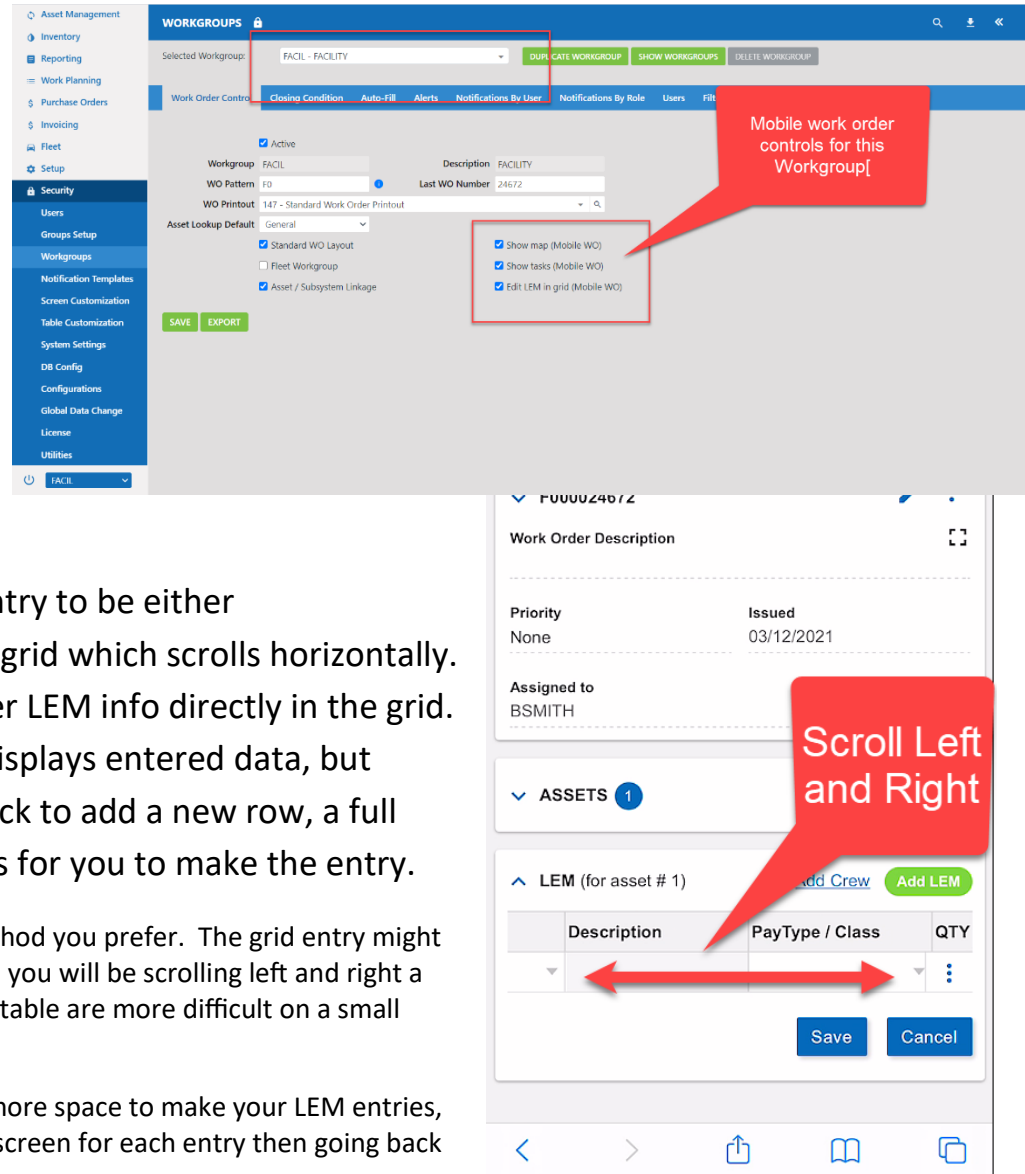
In the Workgroup set up screen in Security, the work order control tab lets you change how your mobile app behaves.

In the work group set up the three check boxes

1. Turn the map on or off
2. Show or not show tasks
3. Change the LEM entry to be either
 - a. A horizontal grid which scrolls horizontally. You can enter LEM info directly in the grid.
 - b. A grid that displays entered data, but when you click to add a new row, a full screen opens for you to make the entry.

It is up to you which display method you prefer. The grid entry might be convenient, but it does mean you will be scrolling left and right a lot, and entries directly into the table are more difficult on a small screen phone.

The full screen entry gives you more space to make your LEM entries, but you will be opening the full screen for each entry then going back to the grid when you save.



Adding a new LEM entry

Click the Add LEM button.

The entry screen will appear. First entry is the type: labor, equipment, contractor, misc., work quantity, or crew. You must make this selection first, as other fields in the line will change depending on this first entry type selection. In this example we will add labor.

The next lines add – activity, management unit, the ID of the craftsman, hours, and minutes, pay type account number, date, location, and charge method.

NOTE: The Management unit and activity codes are now on the labor lines. This means you can now have a work order with multiple management units and Activities. Once you set the Mgt Unit and Activity, the values you select will be automatically carried to the next entries.

Choose the entry values by double clicking on each line. Use the magnifying glass for the craftsman ID. Save when you are done.

The image displays two screenshots of the MAINTSTAR mobile application interface for adding a new LEM entry.

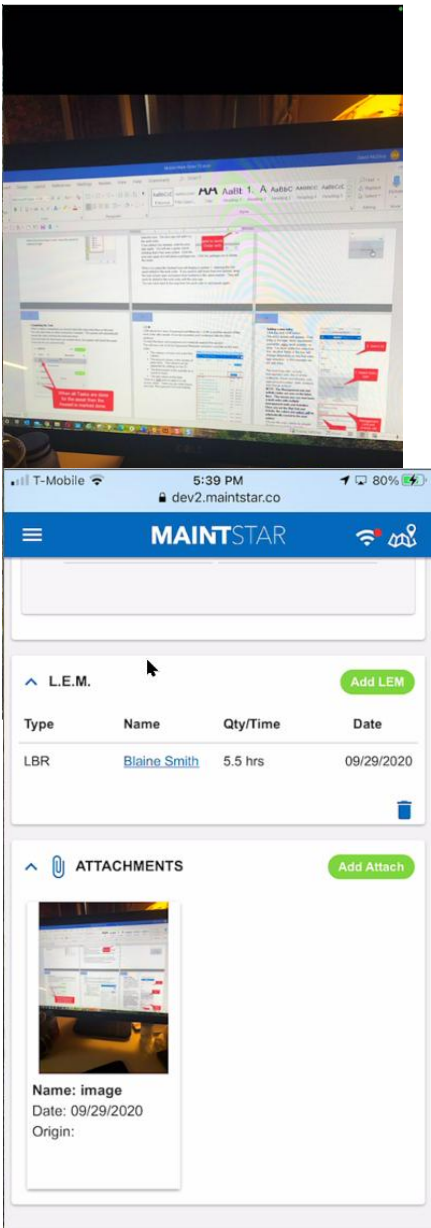
Top Screenshot: The screen is titled "New L.E.M". It features a form with the following fields: "Type *" (set to "Labor"), "ID *" (with a magnifying glass icon), "Description", "Activity", and "Management Unit" (set to "000031 (Utility)"). A red callout bubble labeled "2. Select ID" points to the ID field. Another red callout bubble labeled "1. Select Entry type" points to the "Type *" field.

Bottom Screenshot: The screen shows the same form with additional fields filled out: "Hours *" (set to "5.5"), "PayType / C" (set to "None"), "Account Number", and "Trade". A red callout bubble labeled "Hours in Decimal" points to the "Hours *" field. Another red callout bubble labeled "Management Unit and Activity will carry down to other entries" points to the "Management Unit" field. A keyboard is visible at the bottom of the screen.

Adding Attachments

You can use your phone or tablet to add pictures you take in the field. In the attachment section, click the add attachment button. There you can select a file already on your device or take a new picture. New picture will open your device camera and take the picture. Click use photo. After the photo is taken.

Here you see the Photo attached to the work order.

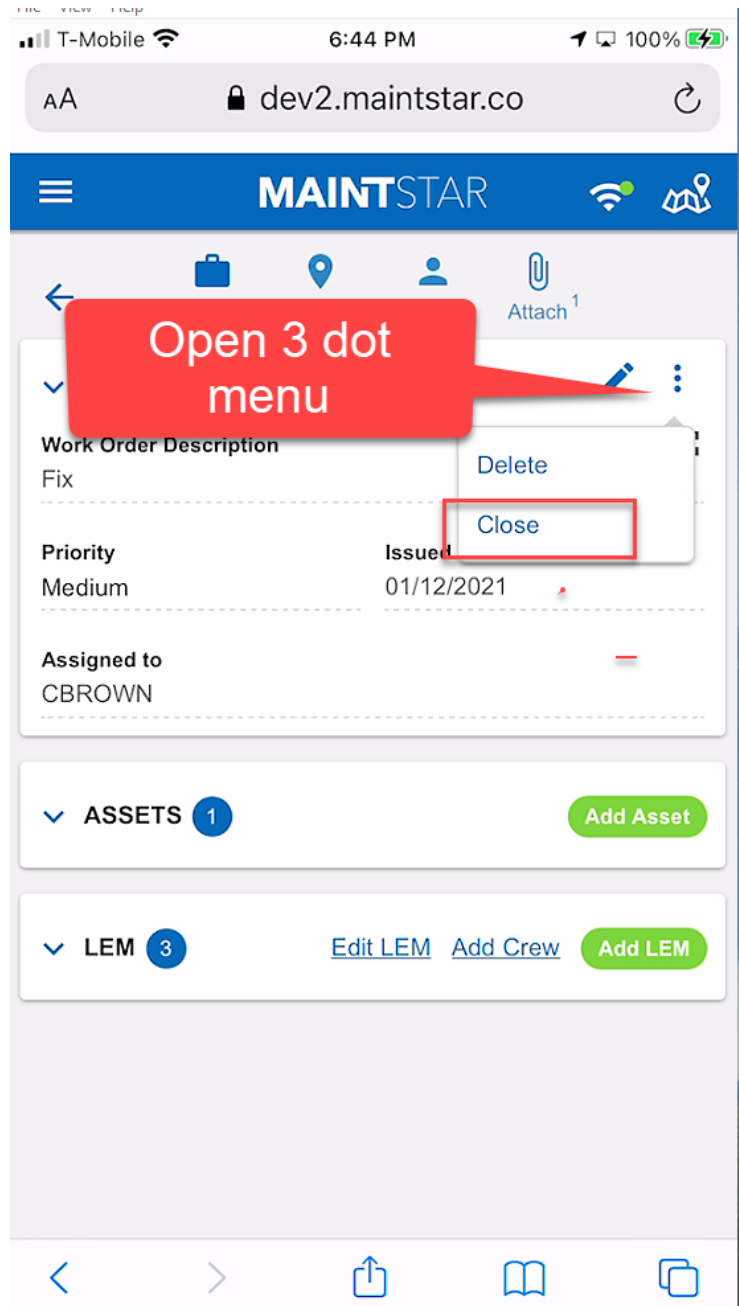


Closing a work order

When you are ready to close the Work Order click on the 3 dot menu on the right top corner of the details screen.

There will be a fly out menu giving a choice of deleting the work order and closing. The delete option will not be functional unless you have delete rights.

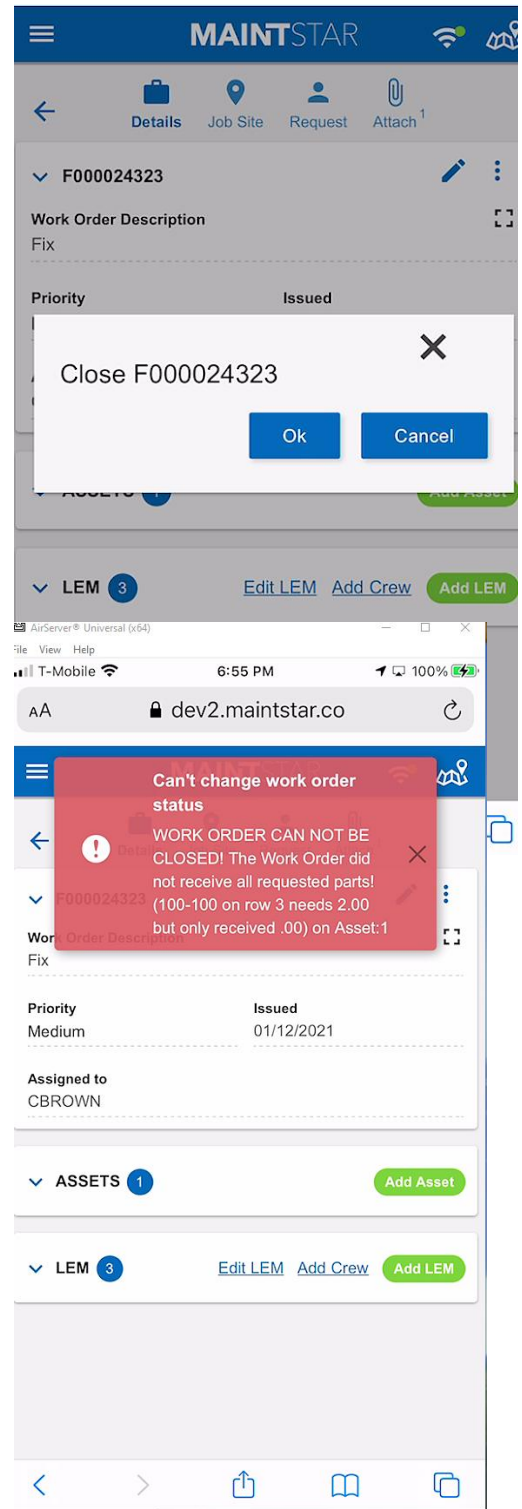
To close the work order, choose 'close'.



The system will ask you to confirm that you want to close.
At this point the system will not close the work order. And the status will be changed to 'closed'

Closing conditions are honored

Please note that the mobile app will honor any closing conditions that have been set for this workgroup.
This means if your administrator has specified that certain fields must be filled in for the work order to close. Here is an example of a work order that had a part entered but the part is from a controlled warehouse that must make an issue transaction for the inventory part before the work order can be closed.
In this case the part has not yet been issued to the work order so the work order cannot be closed.



Actual entry

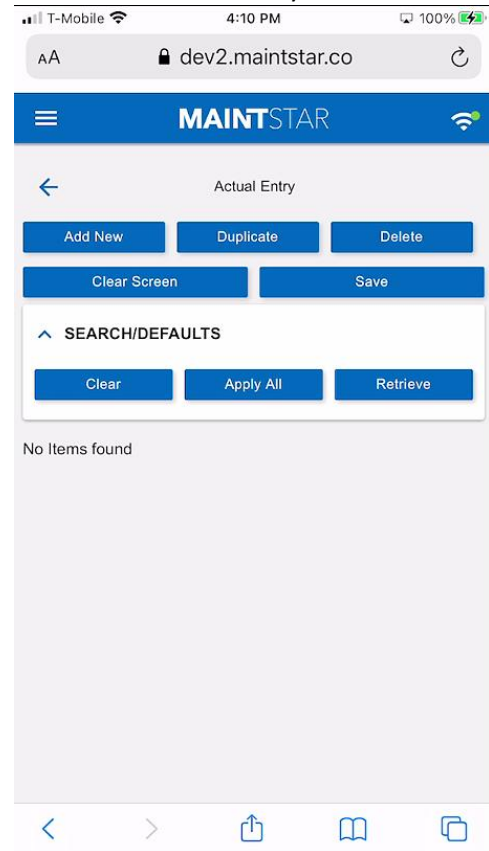
The actual entry section is for making Work Planning entries. This function was previously available only on the desktop. You can now make work planning entries in the field on your portable device.

On the Actual entry screen, you see several blue buttons. Click Add New to start and entry.

- Entry screen
- This is like the desktop entry.
- You first select type of entry
- Labor
- Equipment
- Material
- Contractor
- Miscellaneous
- Or Work Quantity

After type you then select the ID for the entry such as technicians Id.

Other fields will then vary according to the type of entry selected, such as pay type, hours management unit, activity etc. This is very much like the LEM entry on a workorder. Click save then you have the entry completed.

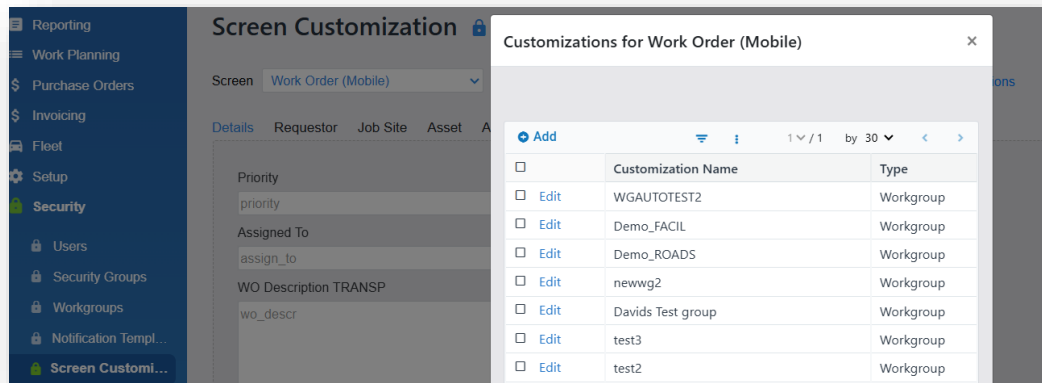


Mobile Work Order Customization

As with desktop screen customization, some mobile work order screens are customizable. In the desktop version, a work order screen customization can be made and then associated with a particular Work Order Type (Setup > Work Order Type > Desktop Customization).

1. To create a Mobile Work Order screen, navigate to Security > Screen Customization. If the intent is to create a unique screen for a particular Work Order Type, click "Add / Remove Customization".

- From this point a customization can be created (or edited).



- The next step is to associate the screen to a Work Order Type.
- Navigate to Setup > Tables > WO Type. Locate or add the desired WO Type.
- Click Edit and specify the desired Customization for Mobile (and/or Desktop)

Table list:
Work Order Type

Work Order Type lookup.

	Work Order Type	Description	Remarks	Color	Inspection	Tv Inspection	Hydrant Inspection	Accident	Customization (Desktop)	Customization (Mobile)
☐ Edit	12345678912345678999				☐	☐	☐	☐		
☐ Edit	ACDT	TRAFFIC ACCIDENT	test		☐	☐	☐	☒		
☐ Edit	ADA	ADA REPAIRS		Light Blue	☐	☐	☐	☐		
☐ Edit	ASPHALT	ASPHALT		Green	☐	☐	☐	☐	TRANSP	
☐ Edit	CALLOUT	STANDBY CALLOUT RESULTING IN O		Purple	☐	☐	☐	☐	WATER	
☐ Edit	CC	Customer Complaint			☐	☐	☐	☐		
☐ Edit	CCTV	Closed Circuit television			☒	☒	☒	☐		
☐ Edit	CIP	CAPITAL IMPROVEMENT PROJECT			☐	☐	☐	☐		
☐ Edit	CM	Corrective Maintenance			☐	☐	☐	☐		
☐ Cancel	CONCRETE	CONCRETE			☐	☐	☐	☐		
☐ Edit	Custom WO	Sara Custom WO			☐	☐	☐	☐	8525	(empty)
☐ Edit	GRAFFITI	GRAFFITI			☐	☐	☐	☐		AIRPRT
☐ Edit	INSP	Inspection		Light Blue	☒	☐	☐	☐		Dauids Test group
☐ Edit	INSP 1	Inspection			☒	☐	☐	☐	8479	Demo_FACIL
☐ Edit	IR	Internal Request			☐	☐	☐	☐		Demo_ROADS
☐ Edit	KT1				☐	☐	☐	☐		ELECT
☐ Edit	KT2				☐	☐	☐	☐		ERM
☐ Edit	LIGHT	STREETLIGHT			☐	☐	☐	☐		FACIL
☐ Edit	Meter Read	Meter Read		Purple	☐	☐	☐	☐		Facility test 2
☐ Edit	MISC	MISCELLANEOUS			☐	☐	☐	☐		FLEET
										LIBRARIES

Row Count: 29